

Experimental Contact I: Exact Nullity Tests

Third-Order Interference and Boost Dispersion in Finite Ring Cosmology

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Abstract

This note executes the two Tier-1 experimental contact points of the FRC quantum programme, both of nullity type, and both closing against data that already exists. *Third-order interference*: because Born weights in the FRC observation formalism are quadratic ledger norms, the Sorkin combinations I_k vanish identically for all $k \geq 3$, on every state, channel, and outcome; below the coherence horizon the carrier shadow reproduces the exact counts, and above it the observable deviation is quantised in integer multiples of the carrier prime. The framework therefore forbids a small nonzero Sorkin parameter at every scale: any statistically confirmed $\kappa \neq 0$, at any precision, falsifies it outright, with no parameter available to absorb the result. Current triple-slit and five-path bounds ($\kappa \lesssim 10^{-3}$) are consistent. *Boost dispersion*: the finite boost-transport identities of the Schrödinger–Dirac companion are verified exhaustively — all 168 boosts of $\mathbb{F}_{13}$, the complete norm-one cycle acting on random spinor fields over $\mathbb{F}_{13}^4$, and full-cycle closure with zero accumulated deformation — so no energy-dependent propagation can arise at any sub-horizon order, and the gamma-ray-burst time-of-flight bounds that exclude generic Planck-suppressed dispersion leave FRC untouched. The closure computation also exhibits the spinor double cover exactly: the frame cycle closes in $(p+1)/2$ steps while the spin lift returns $-I$, requiring two circuits. All claims are machine-verified by exact integer arithmetic (`sorkin.py`, `dispersion.py`; seventeen checks, all passing).

1 Purpose and falsification logic

The FRC quantum formalism [1] reproduces the standard quantum calculus on the sub-horizon class by derivation rather than postulate. Its first experimental obligations are therefore of a particular kind: places where *generic* departures from standard quantum mechanics are experimentally excluded, and where a finite substrate would naively be expected to produce exactly such departures. A theory with tunable parameters survives these tests by retreating; FRC has no tunable parameter at the quantum layer — the carrier scale is fixed elsewhere in the corpus [3, 4] — so each test is a genuine exposure. This note closes the two such tests that require no new theory: the Sorkin hierarchy (Section 2) and boost dispersion (Section 3). Both end in exact null results with a sharp forbidden region, which is the strongest falsifiable posture available to a framework that agrees with quantum mechanics where quantum mechanics has been measured.

2 Third-order interference: the Sorkin hierarchy

Sorkin’s measure-theoretic hierarchy [5] classifies probabilistic theories by the highest order of genuine multi-path interference they admit. For path sets $S \subseteq \{1, \dots, k\}$ with detection weights w_S , the k -th combination is

$$I_k = \sum_{\emptyset \neq S \subseteq \{1, \dots, k\}} (-1)^{k-|S|} w_S. \quad (1)$$

Classical probability has $I_2 = 0$; quantum mechanics has $I_2 \neq 0$ but $I_3 = 0$, a direct expression of the Born rule's quadratic form; theories beyond the quadratic form generically have $I_3 \neq 0$.

Proposition 2.1 (Sorkin nullity on the ledger). *Let $\psi_1, \dots, \psi_k$ be ledger states of an Object in the FRC observation formalism, $\psi_S = \sum_{i \in S} \psi_i$, and let $w_S(r, \alpha) = |\langle v_{r, \alpha}, \psi_S \rangle|^2$ be the outcome weights of [1]. Then I_2 is generically nonzero, and*

$$I_k(r, \alpha) = 0 \quad \text{identically, for every } k \geq 3, \text{ every channel } r, \text{ every outcome } \alpha. \quad (2)$$

Proof. Write $a_i = \langle v_{r, \alpha}, \psi_i \rangle \in \mathbb{Z}[i]$. Sesquilinearity gives $w_S = \sum_{i \in S} |a_i|^2 + \sum_{i \neq j \in S} \bar{a}_i a_j$: a sum of functions each depending on at most *two* slit indices. The combination I_k is the k -fold inclusion-exclusion operator, which annihilates every function of fewer than k slits; for $k \geq 3$ every term of w_S has at most two, so $I_k = 0$. The pair terms survive at $k = 2$, which is interference. *Verified by exact arithmetic on 157/53/13 and 421/61/29 with random Gaussian-integer slit states: $\max |I_2| = 886$ and 840 respectively, while I_3 and I_4 vanish on every channel and outcome (`sorkin.py`).* $\square$

The physical reading is the pair-count identity of [1]: a squared amplitude is a tally over *ordered pairs* of fibre samples. There is nothing in the ledger that counts triples, so there is no third order to interfere. The Born rule's quadratic form, which standard quantum mechanics postulates and experiments such as [6] test, is here the arithmetic of coincidence counting.

Proposition 2.2 (Shadow exactness and wrap quantisation). *Let $\hat{w}_S \in \{0, \dots, p-1\}$ be the carrier shadows of the weights, read through the window lift, and $\hat{I}_3$ the observed Sorkin combination. If all tallies are sub-horizon ($w_S < p$) then $\hat{I}_3 = 0$ exactly. In general*

$$\hat{I}_3 \in p\mathbb{Z}, \quad (3)$$

so the observable third-order interference is either exactly zero or of magnitude at least p .

Proof. $\hat{w}_S = w_S - p[w_S/p]$ and $I_3 = 0$ by Proposition 2.1, so $\hat{I}_3 = -p \sum_S (-1)^{3-|S|} [w_S/p]$. *Verified: with 0/1 coefficients (tallies bounded by $144 < 157$) the observed combination vanishes exactly; with deliberately super-window states, eight nonzero cases appeared, every one a multiple of 157 (`sorkin.py`).* $\square$

Corollary 2.3 (Forbidden region). *FRC admits no small nonzero Sorkin parameter at any tally scale. The normalised κ is either 0 exactly or of order unity (wrap regime); the band in between — precisely where a generic post-quantum theory would sit — is empty.*

Experimental confrontation. Triple-slit photonic experiments bound $\kappa \lesssim 10^{-2}$ [6], improved to the 10^{-3} level by the five-path interferometer of [7]. FRC's prediction at these and all foreseeable scales is $\kappa = 0$ exactly: the wrap regime requires a *single coherent state* whose ledger tally reaches the carrier prime, and the largest laboratory tallies (photon numbers $\sim 10^{20}$ in a pulse, atom numbers $\sim 10^7$ in a condensate) lie tens of orders of magnitude below the coherence horizon $\sqrt{\Omega}$ of the corpus [3, 4]. Three consequences. First, existing null results are passed, and passed identically — not approximately. Second, the Sorkin channel is a *falsification* channel rather than an Ω -measurement channel: improved bounds cannot confirm FRC-specific structure, but a single statistically confirmed $\kappa \neq 0$ at any precision falsifies the framework outright, since Corollary 2.3 leaves it nothing to retreat to. Third, this posture is sharper than that of generalised probabilistic theories, for which κ is a free dial; FRC is the rare finite proposal that *forbids* the deviation its substrate would seem to invite.

3 Boost dispersion: exactness of finite Lorentz transport

Energy-dependent photon velocity is the standard signature of discrete substrates: lattice and automaton models realise the relativistic dispersion relation only approximately, the error accumulates over propagation distance, and gamma-ray-burst time-of-flight measurements now bound the linear-order violation scale beyond the Planck energy [8, 9], excluding the generic case. The FRC Lorentzian layer is not approximate: the Schrödinger–Dirac companion [2] proves the Clifford relations, the spin conjugation formulas, the boost group law, the transported Dirac–Klein–Gordon factorisation, and the exact covariance $D_{\Lambda,S} T_{\Lambda,S} \psi = T_{\Lambda,S} D \psi$ as finite algebraic identities. What this note adds is an independent exhaustive verification, and a closure computation with a physical moral.

The suite (`dispersion.py`) works on the shell $\mathbb{F}_{13}$ with nonsquare $\nu = 2$, gamma matrices over $K = \mathbb{F}_{169}$, and verifies by exact arithmetic: the Clifford relations for all sixteen pairs (D1); the spin conjugation identities for *all* 168 boosts $S(x, y)$ with $x^2 - \nu y^2 \neq 0$ (D2); the boost group law exhaustively on the norm-one cycle (D3); the transported Clifford relations for all boosts (D4); the covariance identity on random spinor fields over the full coordinate shell $Y = \mathbb{F}_{13}^4$, for *every* element of the norm-one boost cycle (D5); the Dirac-to-Klein–Gordon factorisation in the standard and in a boosted frame (D6); and full-cycle closure (D7). Every check passes with zero residual at every lattice point.

Proposition 3.1 (Cycle closure and the finite double cover). *Let $z = x + yc$ generate the norm-one cycle $U_1 \simeq C_{p+1}$ of $K^\times$. The coordinate boost $\Lambda(z)$ has order $(p+1)/2$ in $O(Q_\nu, \mathbb{F}_p)$, while the spin lift satisfies $S(z)^{(p+1)/2} = S(z^{(p+1)/2}) = S(-1) = -I$: transport around the closed boost cycle returns the frame exactly and the spinor with a sign, and only the double circuit restores the identity. Verified at $p = 13$: $\Lambda(3, 2)$ has order 7, $z^7 = -1$, and the composed transport of a random spinor field around the full cycle returns $S^{14} \psi = \psi$ exactly (`dispersion.py`).*

The moral is twofold. First, the spinor double cover — the 2π versus 4π fact of continuum spin — is present in exact finite arithmetic, with the sign living in the two-to-one fibre of $z \mapsto \Lambda(z)$. Second, and decisively for the experimental contact: *nothing accumulates*. A boost orbit traversed completely, step by exact step, returns the field with zero deformation. Dispersion in discrete models is an artefact of approximation error compounding along propagation; in FRC there is no approximation error to compound, at any step, for any element of the boost group, so no energy-dependent velocity can arise at any sub-horizon order. The gamma-ray-burst bounds $E_{\text{QG},1} \gtrsim 7.6 E_{\text{Pl}}$ [8] that falsify generic Planck-scale discreteness therefore do not constrain FRC — a Class-C discrimination: the framework survives, by exactness, a test that eliminates most of its substrate class.

4 Status and next steps

Closed. Both Tier-1 targets are closed as theory-side results: Propositions 2.1–2.2 and Corollary 2.3 (Sorkin), the exhaustive transport verification and Proposition 3.1 (dispersion), all machine-verified — seventeen checks across the two scripts, exact integer arithmetic throughout, no floating point and no limit.

Contributed to the Ω ledger. The Sorkin channel contributes a consistency condition rather than a bound (the wrap regime is inaccessible); the dispersion channel contributes a survival entry: FRC is compatible with time-of-flight bounds that exclude generic discrete substrates. Both entries are parameter-free.

Next. The decisive Tier-2 targets — the CHSH/Tsirelson computation, the network game of the real-versus-complex discrimination, and the gravitationally-induced-entanglement rate — all pass through the ledger-tensor composite construction, scheduled as the programme’s next gate [1].

Reproducibility

The scripts `sorkin.py` and `dispersion.py` reside in the `validation/` folder of the companion draft and are slated for the numeric validation suite at <https://github.com/gamayos/frc-quantum-numeric>, with per-claim mapping to the propositions above. Recorded output: seventeen `PASS` lines and the terminal lines `sorkin: all checks passed`, `dispersion: all checks passed`.

References

- [1] Y. Akhtman. *Quantum Observation in Finite Ring Cosmology: Wave Function as Phase, Amplitude as Subject-Relative Projection, Probability as Coincidence Count*. Companion draft, June 2026.
- [2] Y. Akhtman. *Schrödinger–Dirac Formalism in Finite Ring Continuum*. Preprint, 2026.
- [3] Y. Akhtman and E. Voether. *Gravitation as Phase Synchronisation on a Finite Substrate*. Preprint, 2026.
- [4] Y. Akhtman. *Finite Field Realisation of the Riemann Hypothesis*. Preprint, 2026.
- [5] R. D. Sorkin. Quantum mechanics as quantum measure theory. *Modern Physics Letters A*, 9(33):3119–3127, 1994.
- [6] U. Sinha, C. Couteau, T. Jennewein, R. Laflamme, and G. Weihs. Ruling out multi-order interference in quantum mechanics. *Science*, 329(5990):418–421, 2010.
- [7] T. Kauten, R. Keil, T. Kaufmann, B. Pressl, Č. Brukner, and G. Weihs. Obtaining tight bounds on higher-order interferences with a 5-path interferometer. *New Journal of Physics*, 19:033017, 2017.
- [8] V. Vasileiou et al. Constraints on Lorentz invariance violation from Fermi-Large Area Telescope observations of gamma-ray bursts. *Physical Review D*, 87:122001, 2013.
- [9] A. A. Abdo et al. A limit on the variation of the speed of light arising from quantum gravity effects. *Nature*, 462:331–334, 2009.